

PhD Research Meeting 44

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Advisers: Dr. Simon Jones and Dr. Karin Reinke

1. Reframing the RQ2 Punchline
2. Baseline Characterization
3. STL Decomposition and BAU Projection



RQ2 punch line: when does NTL recovery agree with physical recovery, and what explains the mismatch?

1. Characterise functional recovery from NTL

Shock → trajectory → recovery metrics / response families

2. Match it with physical EO recovery

Building damage • reconstruction • land-cover / built-up change

same location • comparable spatial support • aligned recovery period

3. Measure agreement and divergence

- Where does NTL recovery track physical reconstruction?
- Where does one recover faster or slower than the other?
- How large and spatially persistent are those differences?

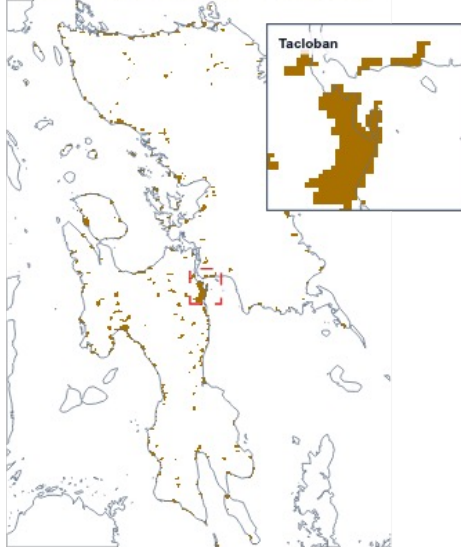
4. Explain the divergence using mapped context

Settlement class • land-cover type • population density / redistribution • exposure • urban morphology

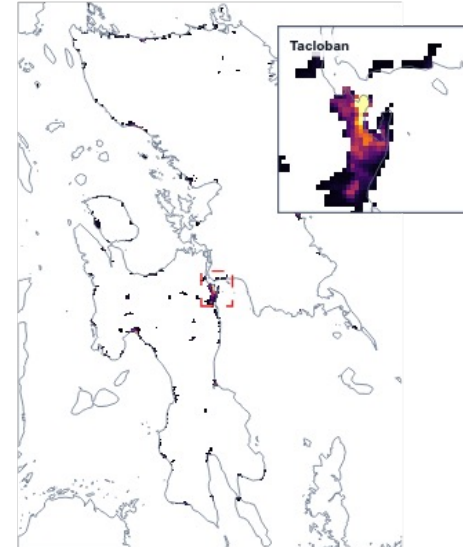
How reliably can nighttime lights reproduce the spatial and temporal pattern of physical post-disaster recovery, and where they diverge, what mapped characteristics explain why?



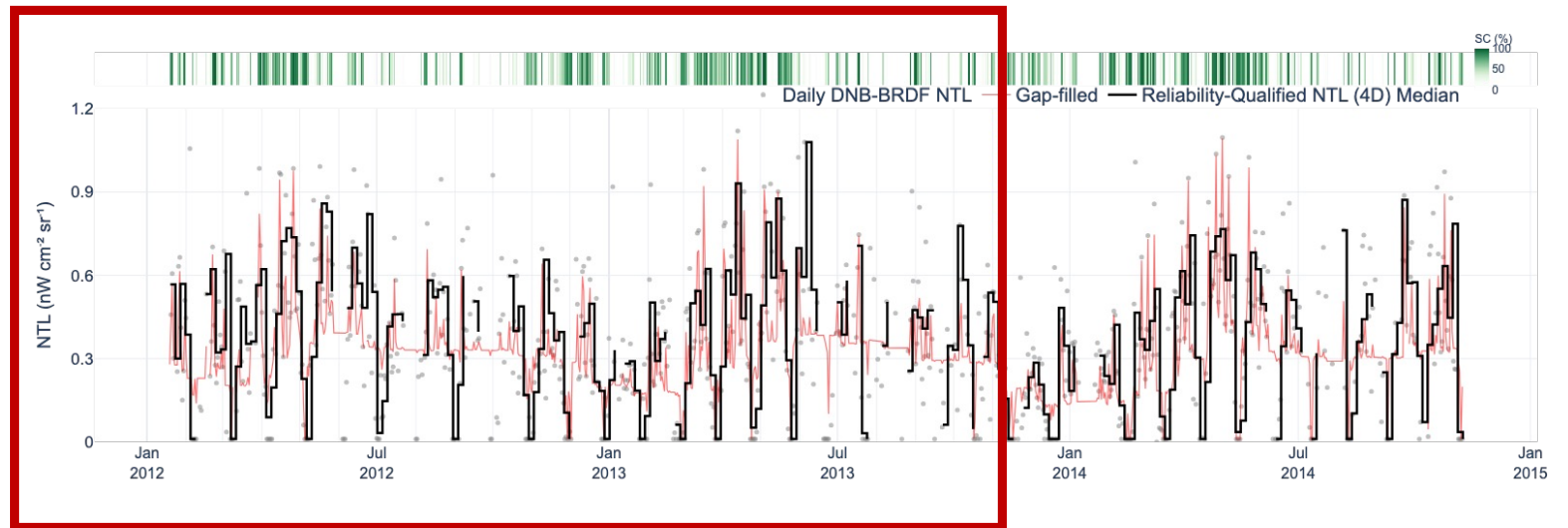
c. G3 settlement mask
classes 22, 23 and 30 · 4,452 pixels

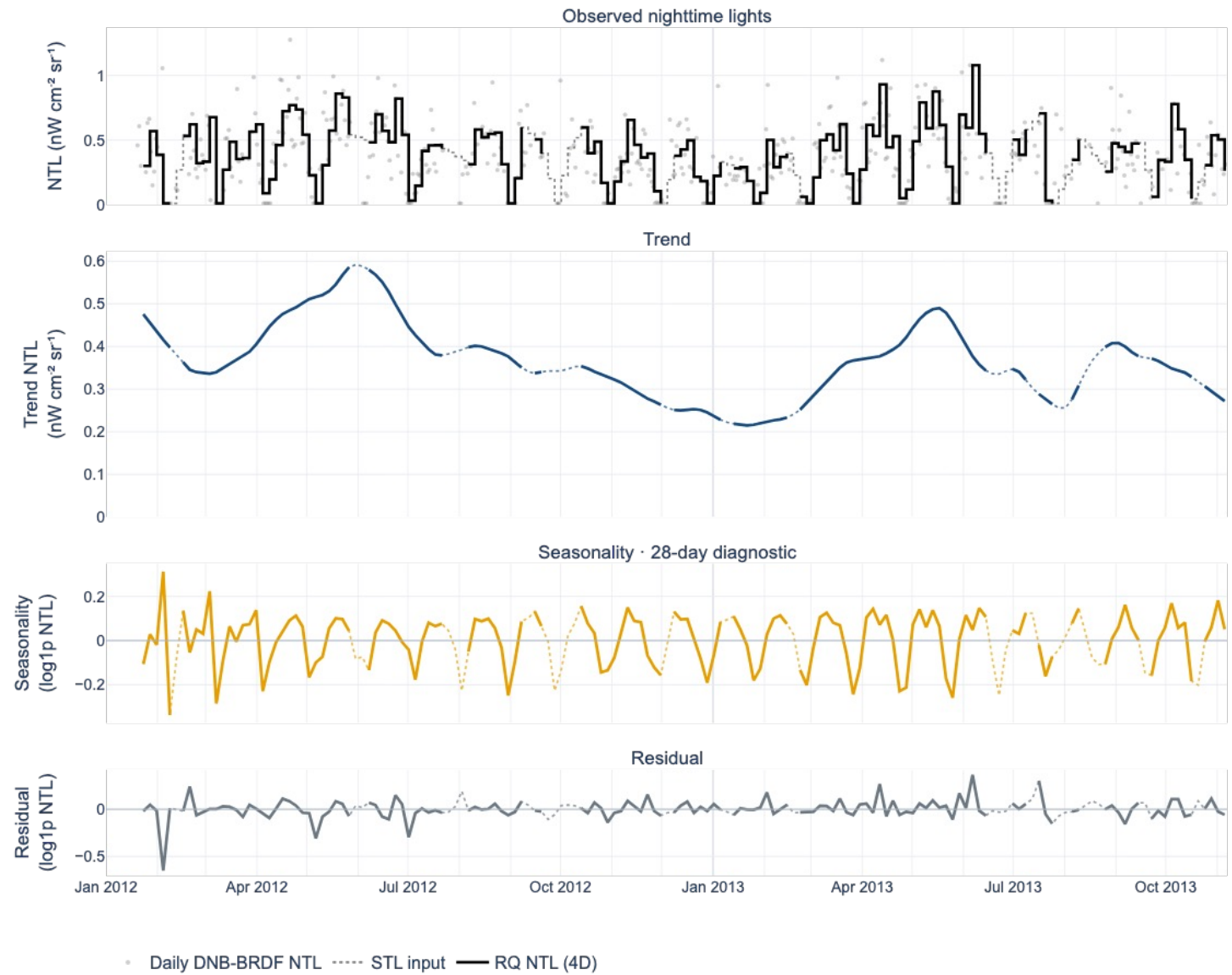


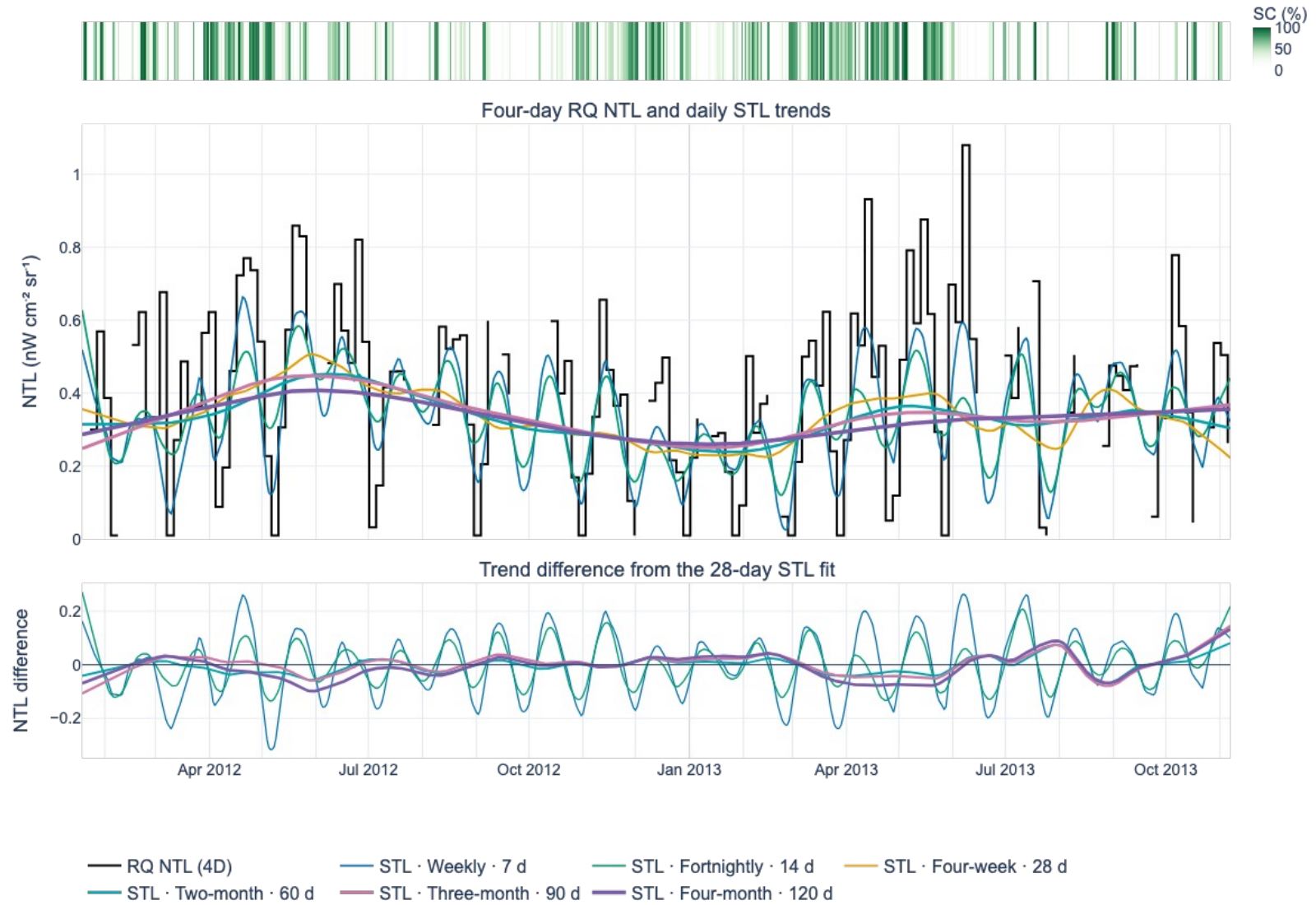
f. Pre-Haiyan median DNB-BRDF
within G3

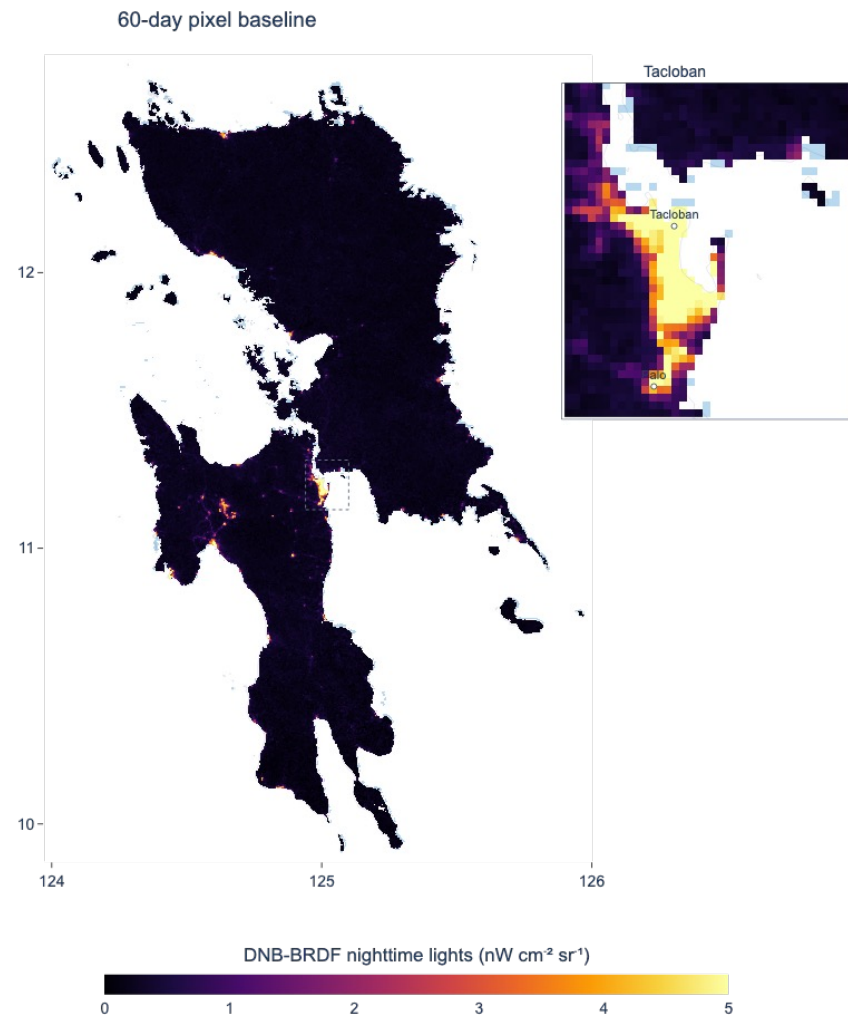
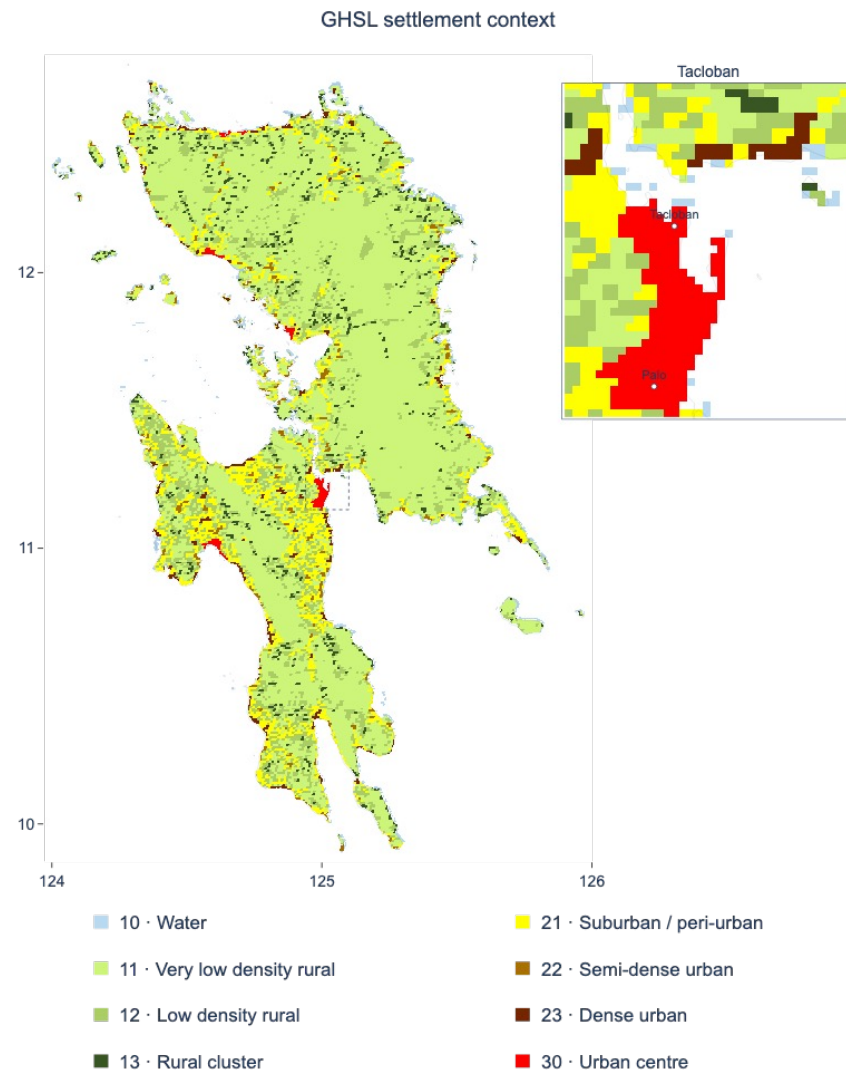


Median radiance
($\text{nW cm}^{-2} \text{sr}^{-1}$)

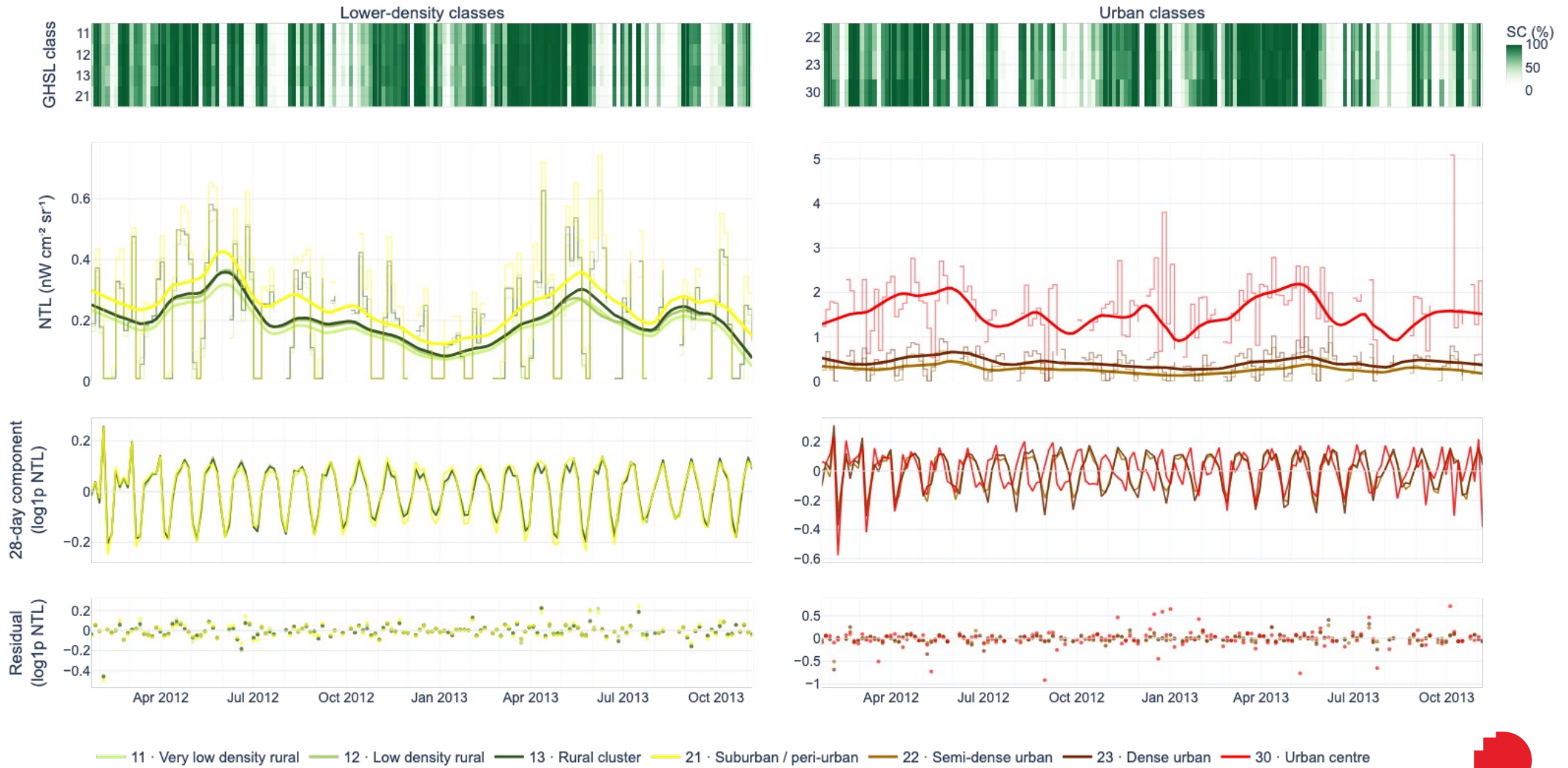


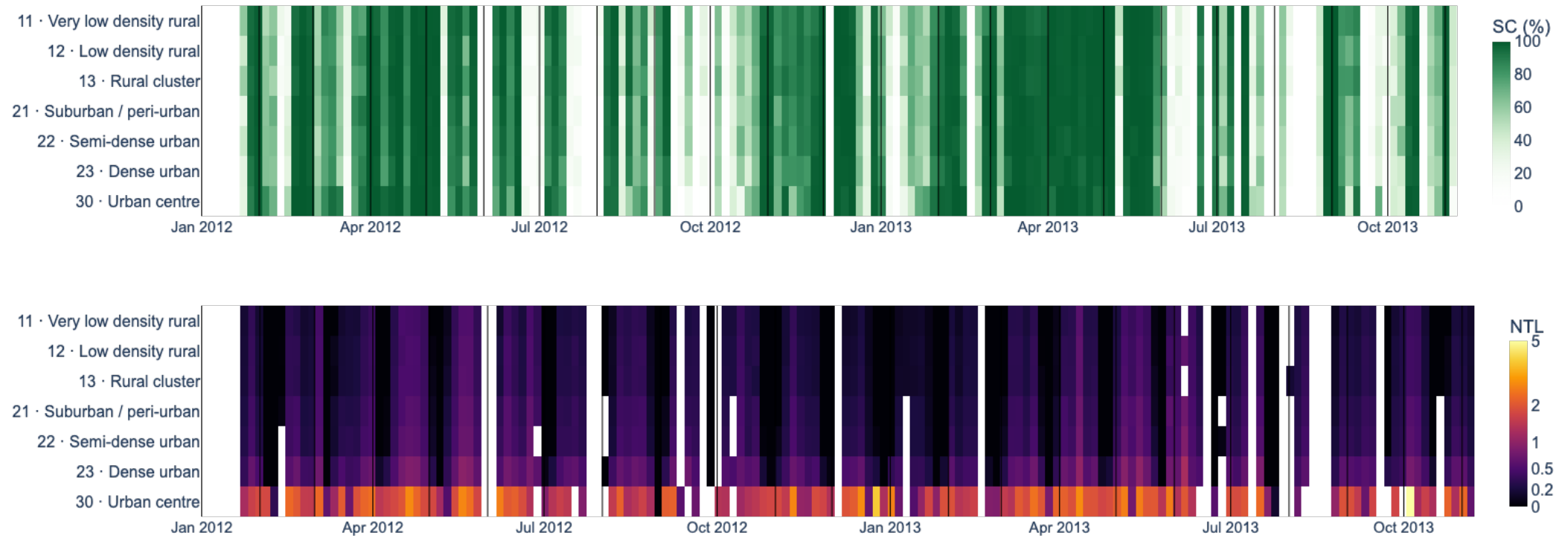


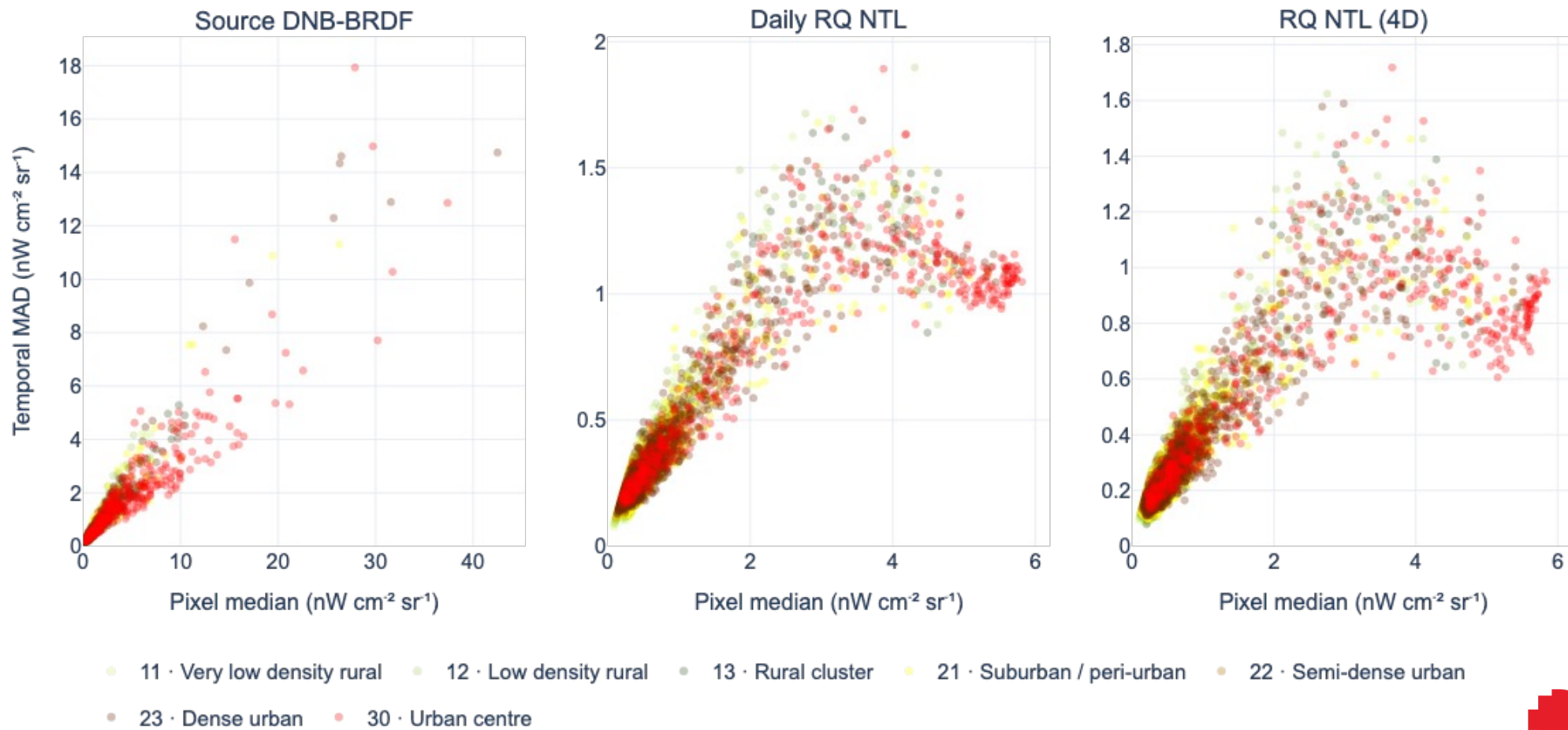




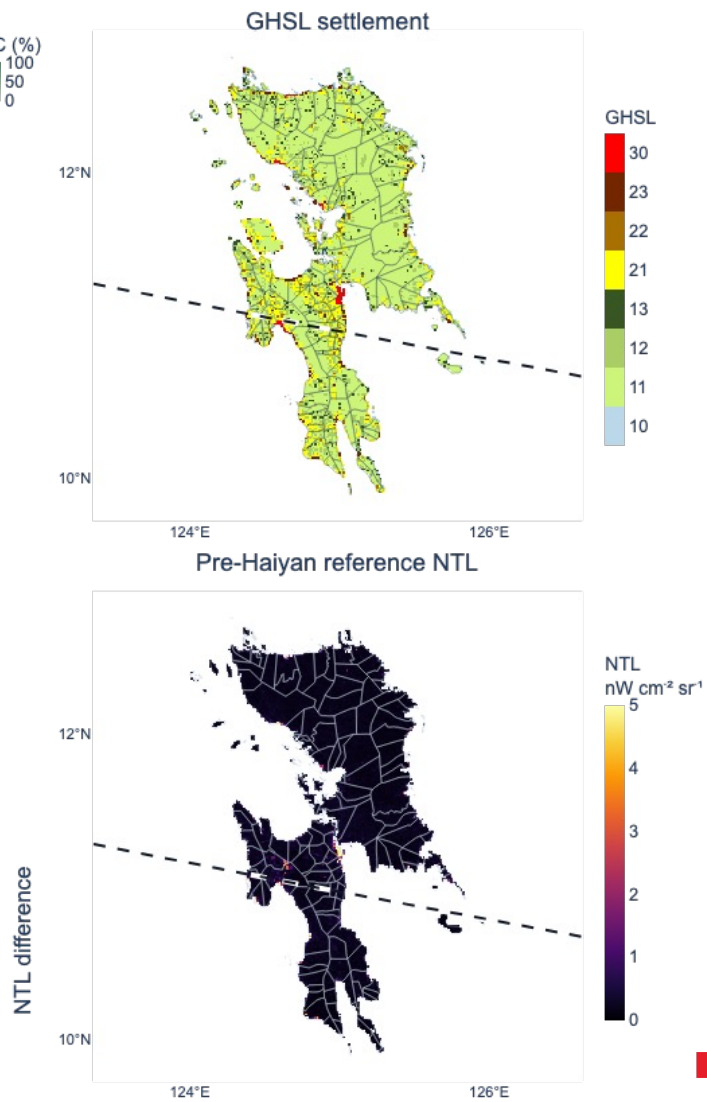
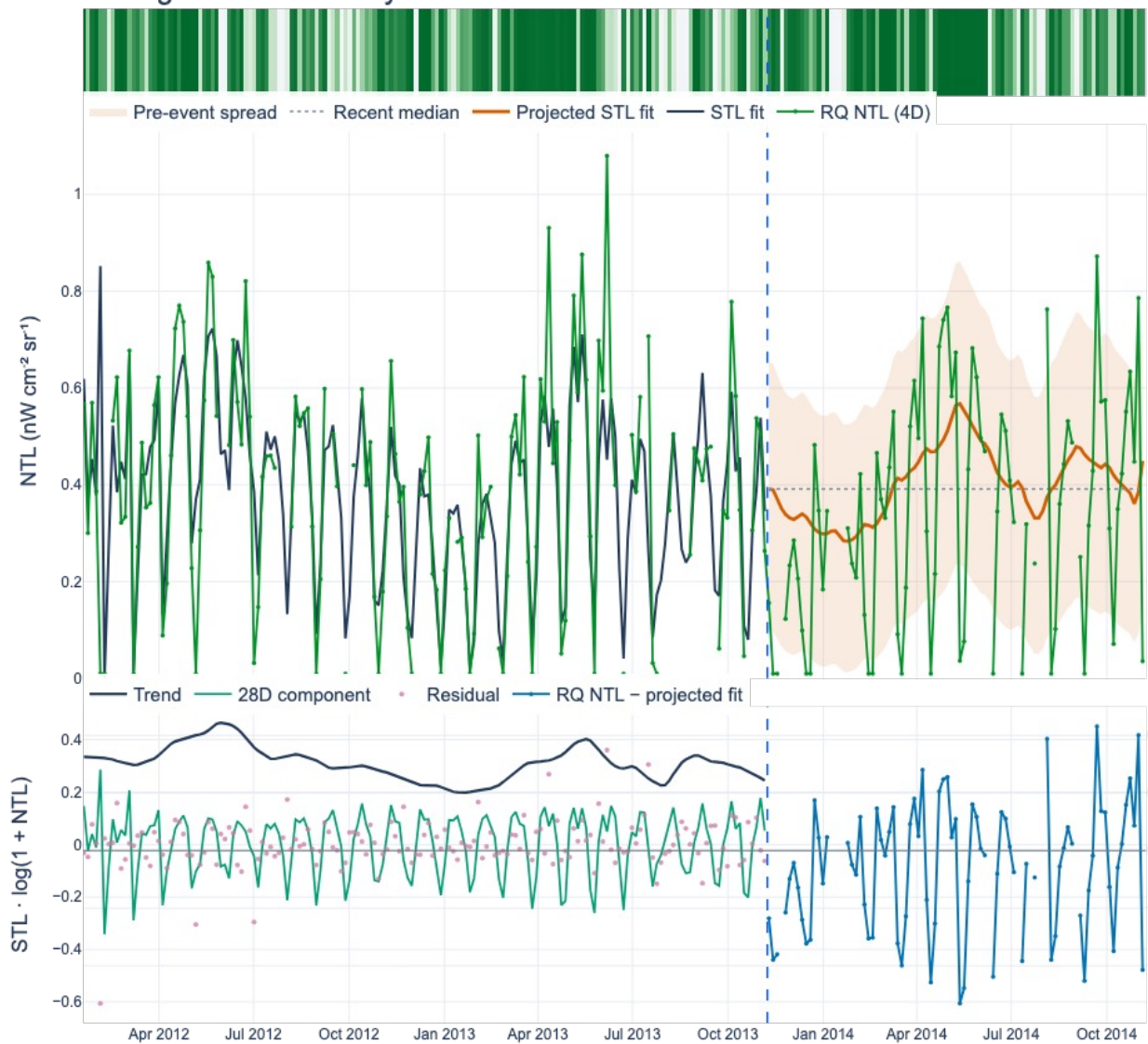




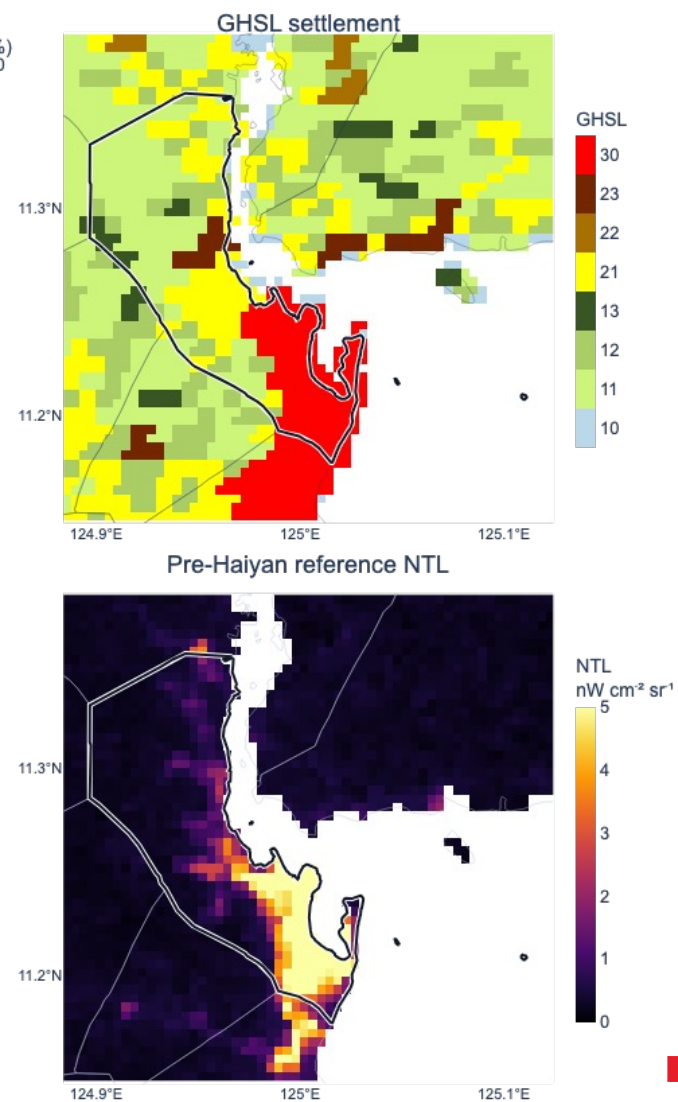




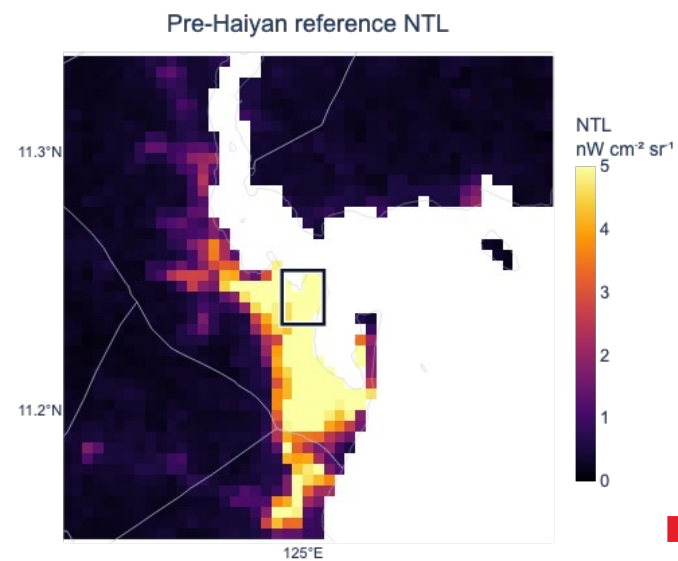
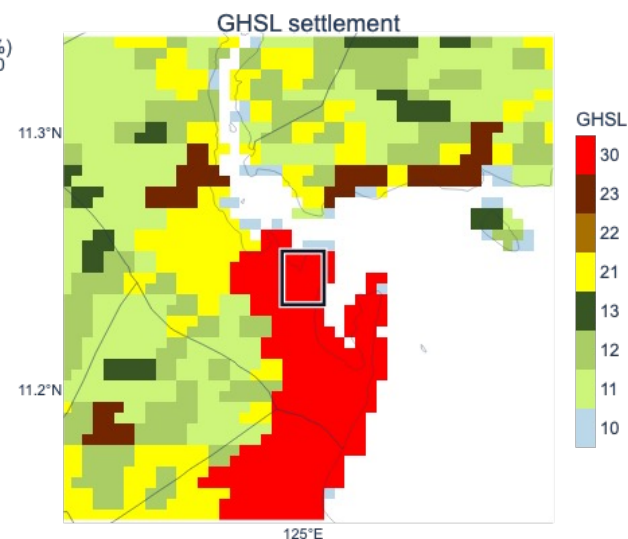
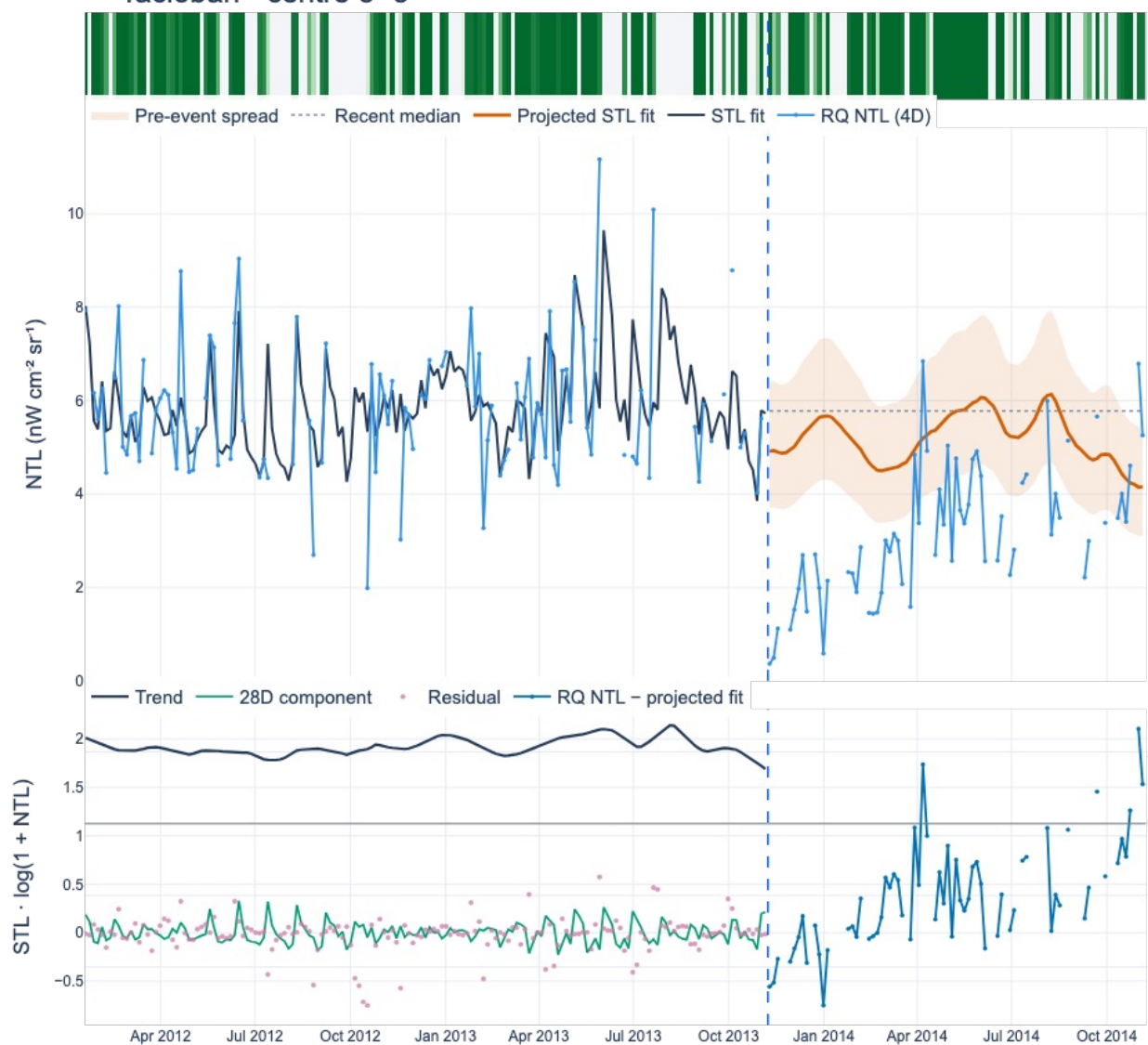
Regional Samar-Leyte G3



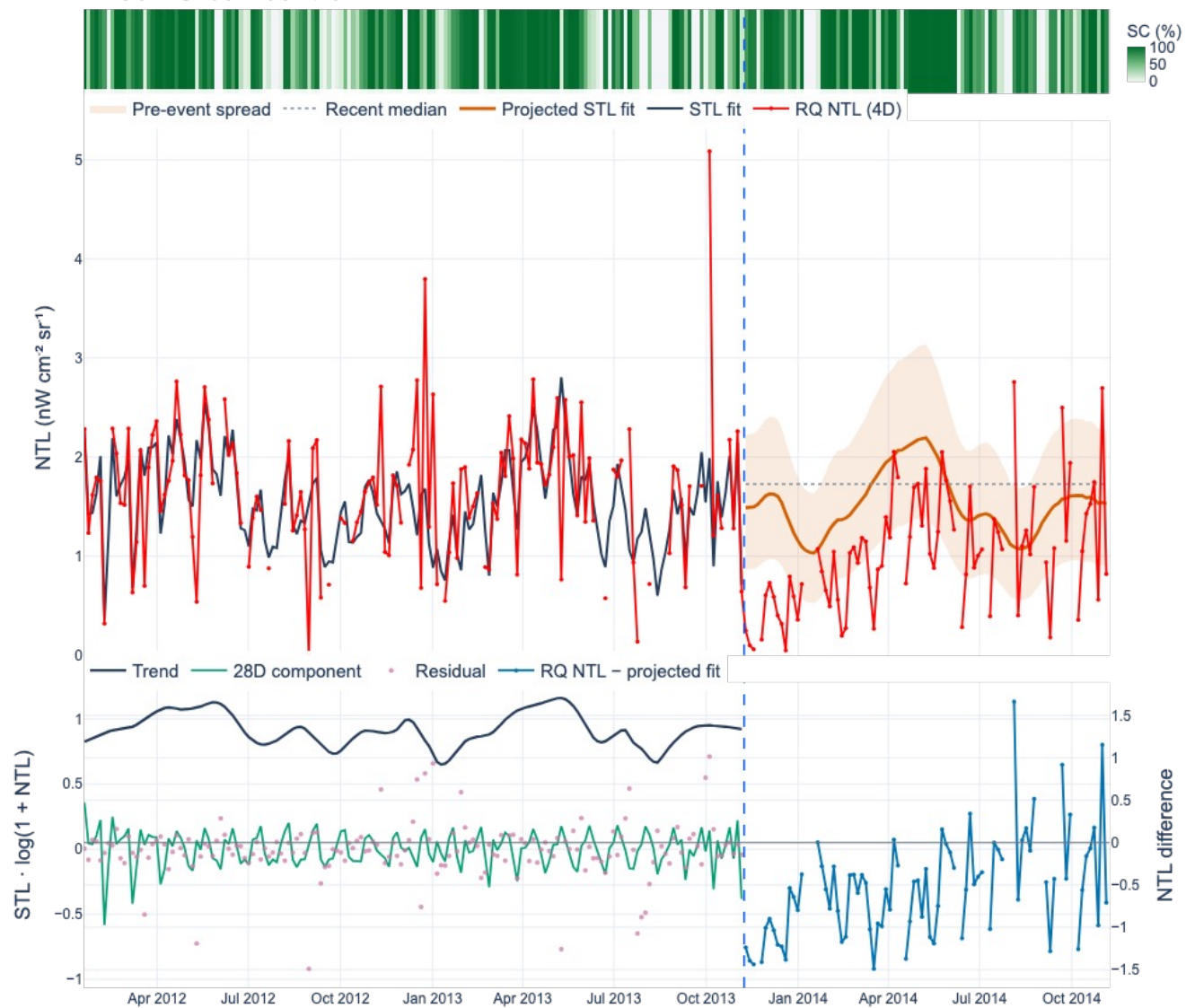
Tacloban



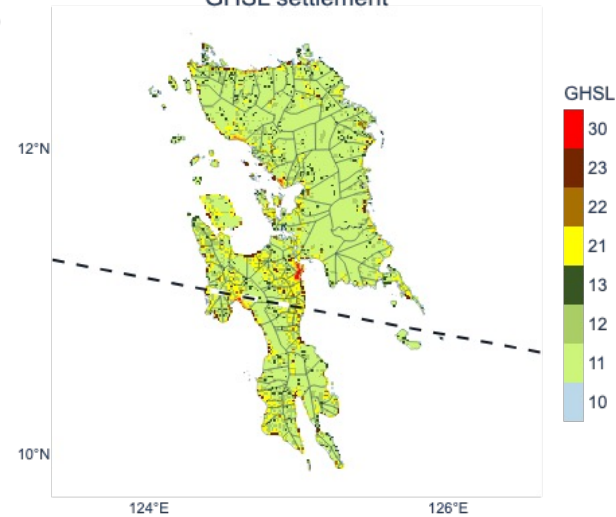
Tacloban · centre 5×5



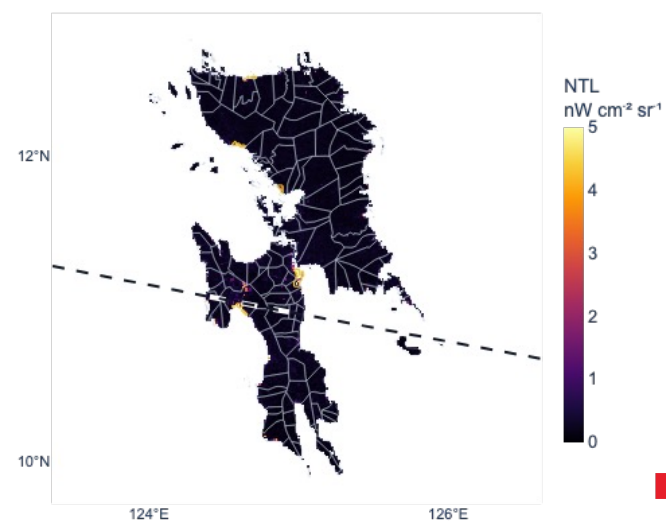
30 · Urban centre

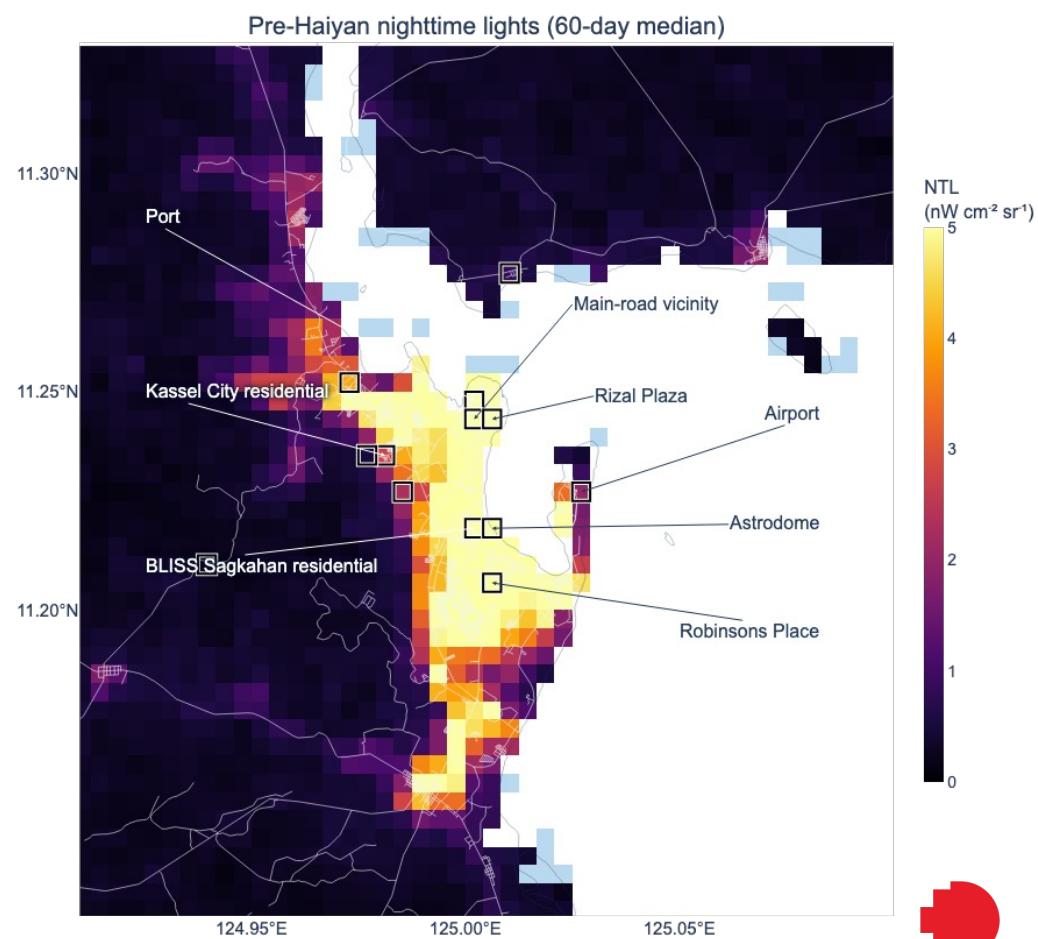
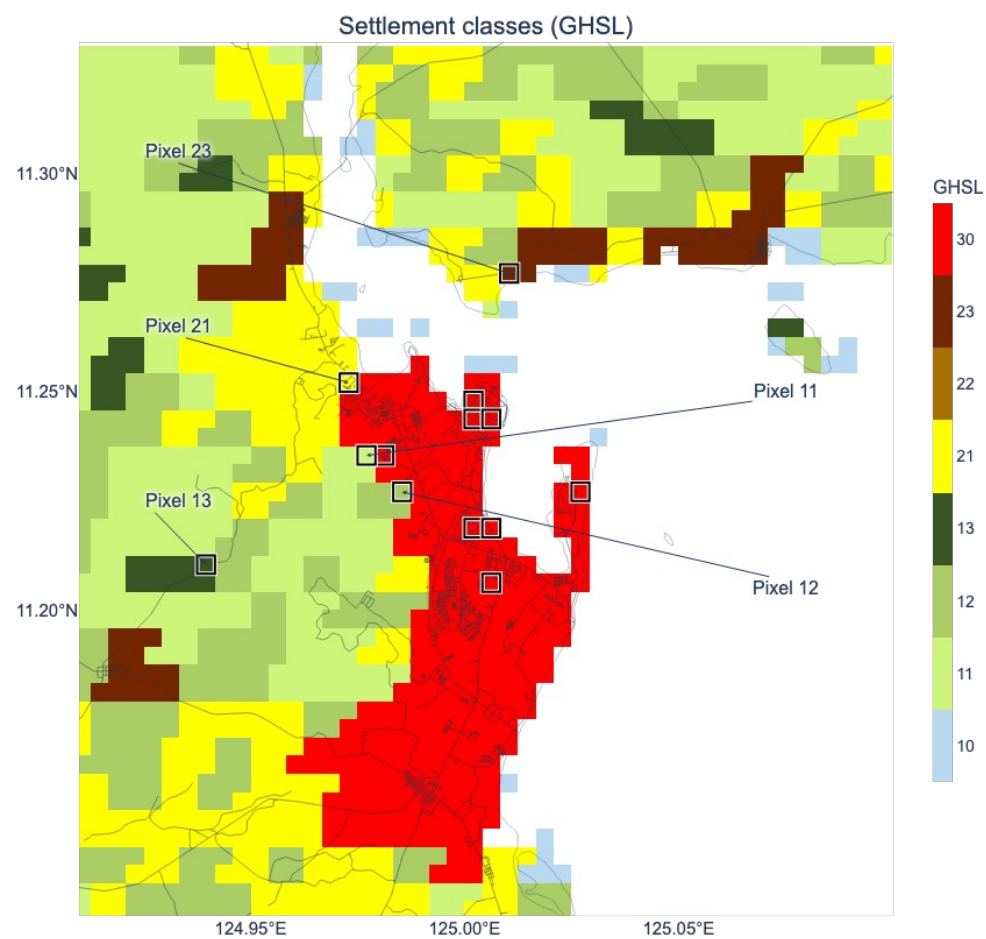


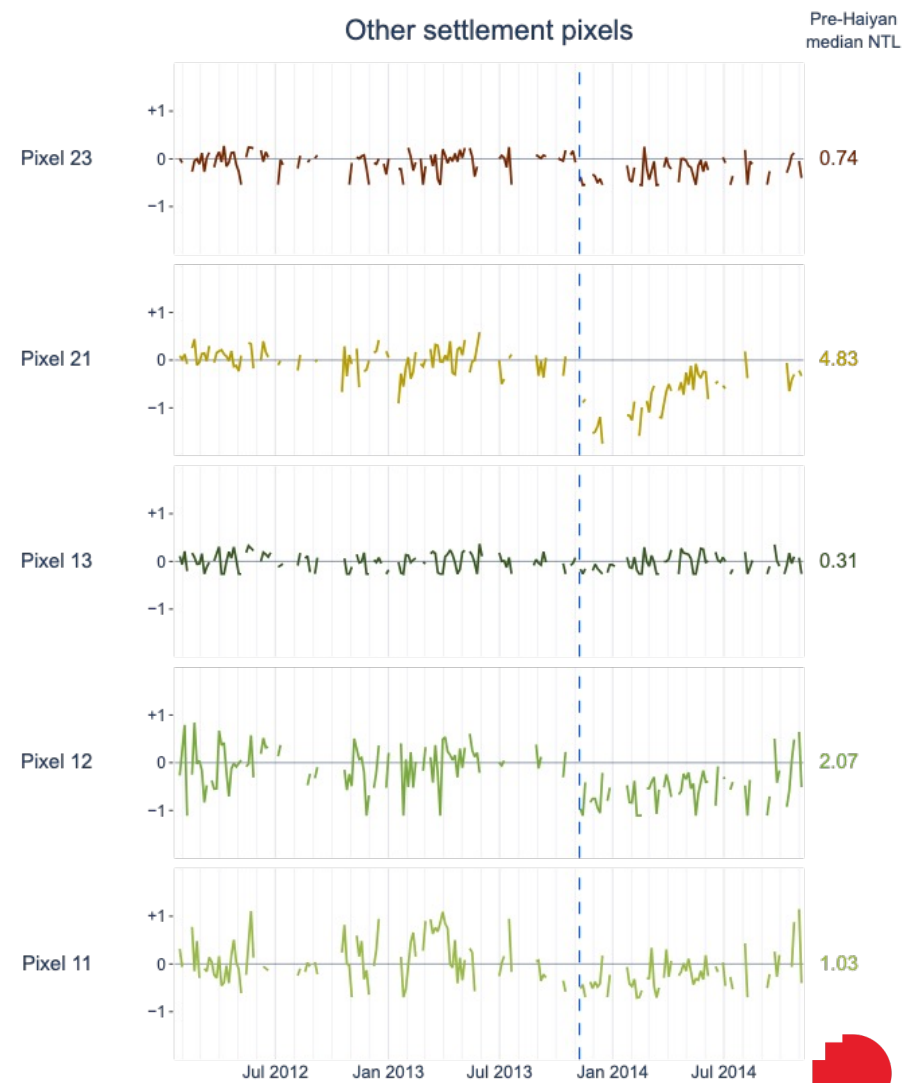
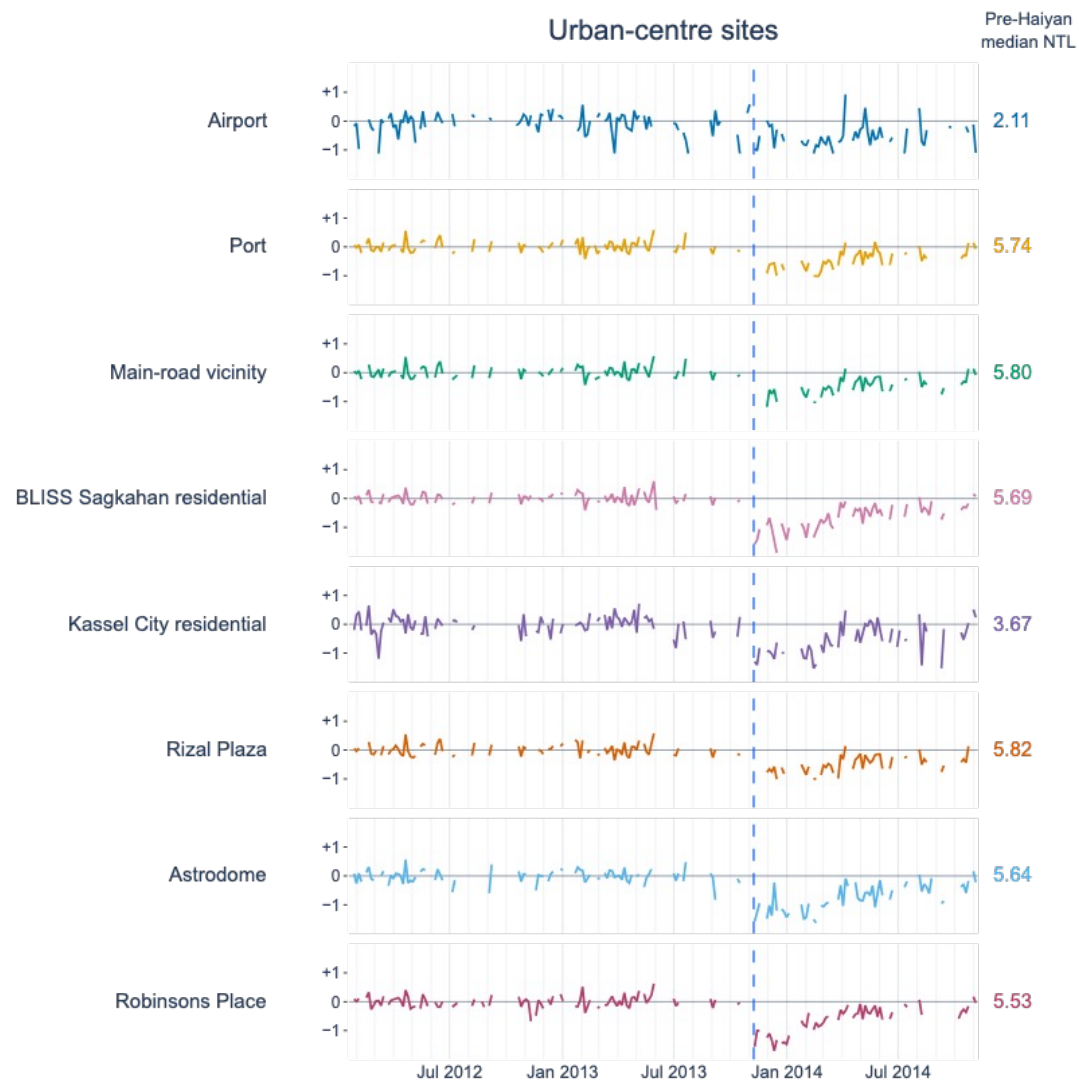
GHSL settlement

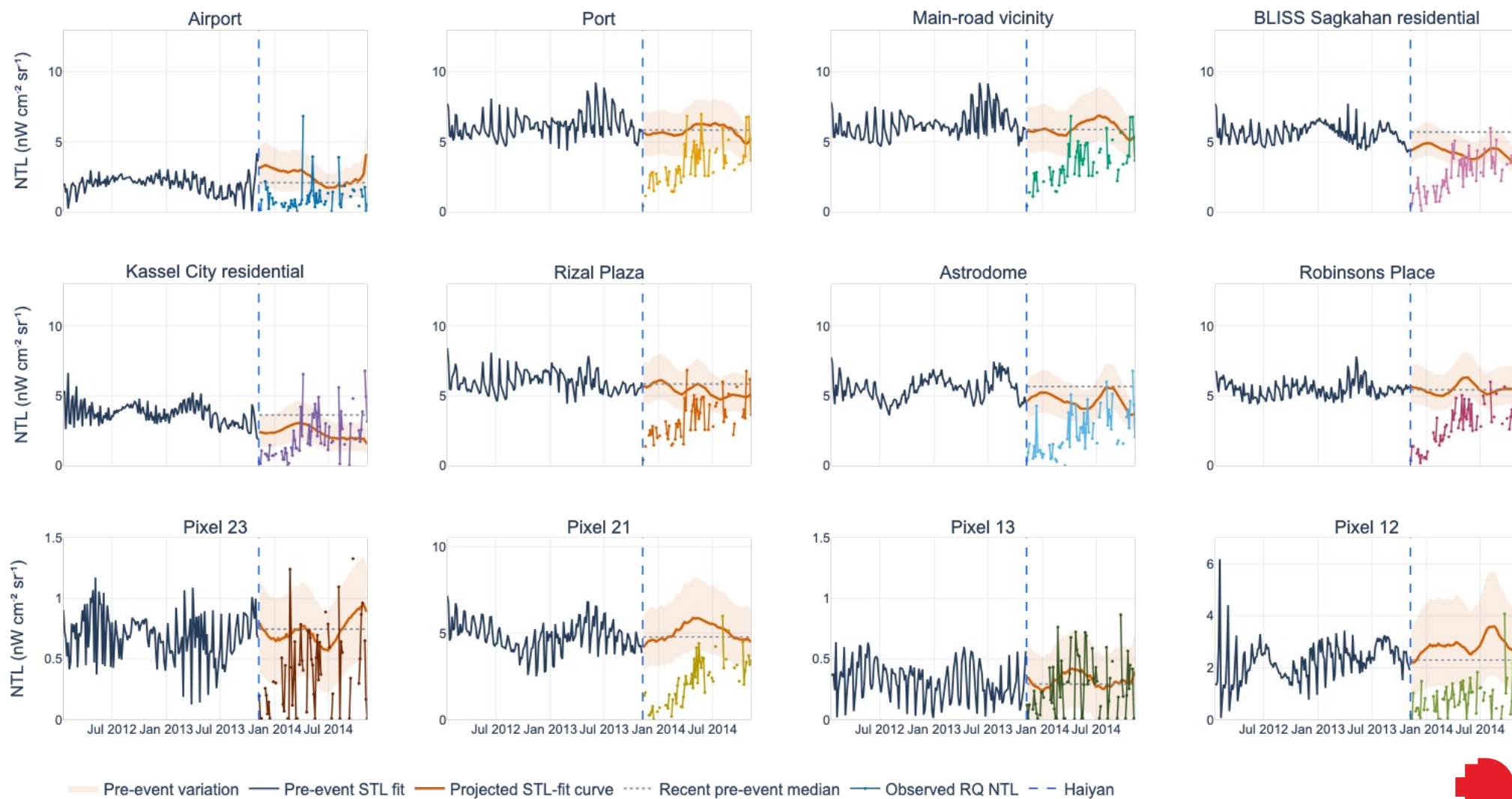


Pre-Haiyan reference NTL









Airport

